FEDERAL STANDARD STOCK CATALOGUE

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

GLYCERIN (GLYCEROL)

This specification was approved for promulgation by the Federal Specifications Board on February 3, 1931, for the use of the departments and independent establishments of the Government in the purchase of this commodity and shall become mandatory for all departments and independent establishments of the Government not later than August 1, 1931. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE FEDERAL SPECIFICATIONS.

A–1. There are no Federal specifications applicable to this specification.

A–2. Any special requirements of the individual departments of the Government are noted under section H.

B. GRADES.

B–1. Glycerin shall be of the following grades, as specified:

Grade A—"U. S. P."
Grade B—"High-gravity" ("Dynamite glycerin").
Grade C—"Yellow distilled."

C. MATERIAL.

C–1. Shall be as specified hereinafter for each grade.

D. GENERAL REQUIREMENTS.

D–1. See section E.

E. DETAIL REQUIREMENTS.

E–1. *Grade A.*—"U. S. P." glycerin shall conform to the requirements of the latest revision of the United States Pharmacopœia for glycerin.

E–2. *Grade B.*—"High-gravity" glycerin ("Dynamite glycerin").

E–2a. *Appearance.*—Shall be clear and free from suspended matter.

Source: https://assist.dla.mil -
Check the source to verify that t

E–2b. *Color.*—Maximum limit shall not be greater than a combination of No. 80 yellow Lovibond glass and No. 20 red Lovibond glass on a 5¼-inch column of the glycerin.

E–2c. *Odor.*—Shall be slight and characteristic of the grade of glycerin specified. If desired, shall conform to the odor of a sample mutually agreed upon by buyer and seller.

E–2d. *Specific gravity.*—Shall be not less than 1.2620 at 15.5/15.5° C.

E–2e. *Acidity or alkalinity.*—When 50 ml. of the glycerin is mixed with 100 ml. of distilled, carbon dioxide-free water and about 0.5 ml. of phenolphthalein indicator, the solution shall not require more than 0.30 ml. of normal hydrochloric acid solution or more than 0.30 ml. of normal sodium hydroxide solution for neutralization.

E–2f. *Ash.*—Shall not exceed 0.10 per cent.

E–2g. *Chlorides, calculated as chlorine.*—Shall nor exceed 0.01 per cent when determined volumetrically in an aqueous solution of the residue left in the determination of the ash content.

E–3. *Grade C.*—"Yellow-distilled" glycerin.

E–3a. *Appearance.*—Shall be clear and free from suspended matter.

E–3b. *Color.*—Maximum limit shall not be greater than a combination of No. 80 yellow Lovibond glass and No. 20 red Lovibond glass on a 5¼-inch column of the glycerin.

E–3c. *Odor.*—Shall not be disagreeable. If desired, shall conform to the odor of a sample mutually agreed upon by buyer and seller.

E–3d. *Specific gravity.*—Shall be not less than 1.2550 at 15.5/15.5° C.

E–3e. *Acidity or alkalinity.*—When 50 ml. of the glycerin is mixed with 100 ml. of distilled, carbon dioxide-free water and about 0.5 ml. of phenolphthalein indicator, the solution shall not require more than 0.30 ml. of normal hydrochloric acid solution or more than 0.30 ml. of normal sodium hydroxide solution for neutralization.

E–3f. *Ash.*—Shall not exceed 0.02 per cent.

E–3g. *Chlorides, calculated as chlorine.*—Shall not exceed 0.01 per cent when determined volumetrically in an aqueous solution of the residue left in the determination of the ash content.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Sampling.*—Glycerin being very hygroscopic, the samples should be taken, if possible, in clear, dry weather.

F–1a. *Barrels and drums.*—Not less than 20 per cent of the packages in any shipment or delivery shall be represented in the sample. Thoroughly mix the contents of each barrel or drum to be sampled by rolling the container, and withdraw a portion from about the center by means of a clean, dry "thief" or a suitable glass tube, lowering the sampling device very gradually into the glycerin. The "thief" or tube shall be stripped as it is drawn from the barrel or drum, of the outer coating of glycerin, by pulling it through the hand (or a clean, dry cloth) while tightly clasped around it and allowing the drippings to flow into a waste can. When changing from one barrel or drum to another, the "thief" or sampling tube shall be rinsed at least twice, in the same way the sample is taken, in the glycerin of the new barrel or drum, letting the glycerin flow into a waste can. The sample drawn, as above described, from each barrel or drum

Source: https://assist.dla.mil
Check the source to verify that

shall be quickly transferred to a *clean, dry,* 16-ounce glass bottle and tightly stoppered with a clean, dry stopper, after wiping any glycerin from the neck of the bottle with a clean, dry cloth. A separate bottle shall be used for each sample drawn, and each bottle should be labeled to show the container from which its contents were taken, the date of sampling, and the name of the inspector. One-third of the contents of each glass bottle shall be poured into each of three clean, dry, clear glass bottles of suitable capacity, thus forming composite samples of the lot, tightly stoppered with a new, clean, dry cork or a glass stopper, and sealed. These composite samples shall be mixed by shaking, after standing the bottles in warm water until the glycerin flows freely. The bottles shall then be wiped off and distinctly labeled by the inspector. One composite sample shall be delivered to the buyer, one to the seller if he so requests, and the third held for check in case of dispute. All acceptance tests shall be performed on a composite sample.

F–1b. *Small containers, cans, etc., of 10 gallons or less.*—Thoroughly mix the contents of each package to be sampled by shaking or inverting several times. Samples shall be drawn by means of a glass tube or "thief," as described in F–1a from at least five packages (from all when fewer), and in no case from less than 10 per cent of the packages. These samples shall be handled and composite samples prepared as described in F–1a.

F–1c. *From loaded tank car or other large vessel.*—The composite sample taken shall be not less than 5 gallons, and shall consist of numerous small samples taken from various points by means of a suitable tube or "thief," as described in F–1a. The composite sample thus obtained shall be thoroughly mixed, and from it three samples of not less than 1 quart each shall be placed in clean, dry, glass bottles, which must be nearly filled with the sample and securely stoppered with new, clean, dry corks or tightly fitting glass stoppers. These shall be sealed and distinctly labeled by the inspector; one shall be delivered to the buyer, one to the seller, and the third held for check in case of dispute.

F–1d. *While loading tank car or while filling containers for shipment.*—This method shall be used as an alternative method and only when mutually agreed upon by buyer and seller. Samples shall be drawn by the purchaser's inspector or other sampler mutually agreed upon by buyer and seller at the discharge pipe where it enters the receiving vessel or vessels, or shall be continuously withdrawn by means of a "bleeder" valve in the pump riser. The composite sample shall be not less than 5 gallons and shall consist of small portions of not more than 1 quart each taken at regular intervals during the entire period of loading or filling, or shall consist of the sample continuously withdrawn by means of a "bleeder" valve in the pump riser. The composite sample thus obtained is handled as in F–1c.

F–2. *Testing.*

F–2a. *Appearance.*—Examine the thoroughly mixed sample, free from air bubbles, in the container in which submitted (or a portion in a test tube) by transmitted light to determine compliance with the specifications.

Source: https://assist.dla.mil
Check the source to verify that

F–2b. *Color.*—Compare the depth of color of a 5¼-inch column of the sample with that given by the combination of Lovibond glasses specified for the grade of glycerin under examination. This may be done by placing the sample to a depth of 5¼ inches in a colorimeter tube which is then put in a colorimeter; under or over a similar empty colorimeter tube placed in the colorimeter, put the Lovibond glasses, and compare the colors. Instead of colorimeter tubes, 4-ounce oil sample-bottles of colorless glass with flat, polished bottoms and with square shoulders may be used. In cases of dispute the Lovibond glasses used shall be a set standardized by the United States Bureau of Standards.

F–2c. *Odor.*—Note whether the odor is slight, marked, or disagreeable. Determine by comparison with the agreed-upon sample.

F–2d. *Specific gravity.*—Determine at the specified temperature in comparison with water at the same temperature by any method that is accurate within two points in the fourth decimal place.

F–2e. *Acidity or alkalinity.*—Measure 50 ml. of the sample in a 50 ml. graduated cylinder and pour into a 200-ml. Erlenmeyer flask. Rinse the graduated cylinder, with 100 ml. of freshly boiled and cooled (carbon-dioxide-free) distilled water that has been rendered exactly neutral to phenolphthalein indicator, pouring the rinsings into the Erlenmeyer flask containing the sample. Titrate the glycerin solution to neutrality with 0.5 *N* HCl solution or 0.5 *N* NaOH solution as required, using phenolphthalein indicator. Multiply the number of milliliters required by the normality of the solution used and report the product as milliliters of normal solution required for 50 ml. of the sample.

F–2f. *Ash.*—Heat about 50 g. (either weighed directly or a definite volume multiplied by the specific gravity) of the sample in a weighed dish over a free flame until the vapor continues to burn after removing the flame. Carefully protect the operation from drafts. When the combustion dies out, ignite the residue at a low red heat until the carbonaceous matter is consumed. Cool in a desiccator and weigh. Calculate the percentage of ash.

F–2g. *Chlorides, calculated as chlorine.*—Add about 25 ml. of hot distilled water to the ash residue obtained in F–2f and rub with the flattened end of a stirring rod. Add 1 ml. of potassium chromate indicator and titrate with 0.01 *N* $AgNO_3$ solution to the first permanent reddish tint. Calculate the percentage of chlorides as chlorine based on the weight of sample used, F–2f. (1 ml. 0.01 *N* $AgNO_3$=0.0003546 g. Cl.)

F–3. *Reagents.*

F–3a. *Standard sodium hydroxide solution (approximately 0.5 N).*—Prepare a saturated solution of NaOH in freshly boiled distilled water in a stoppered flask; let settle overnight, and transfer a clear 35-ml. portion to a 1-liter volumetric flask. Dilute to 1,000 ml. with recently boiled distilled water and mix, cool to room temperature, and standardize against Bureau of Standards standard acid potassium phthalate, using phenolphthalein indicator. Dilute, if desirable, restandardize, and protect the solution from CO_2.

Source: https://assist.dla.mil -- Check the source to verify that thi

F–3b. *Standard hydrochloric acid solution (approximately 0.5 N).*—To pure HCl (specific gravity 1.19) add distilled water until the specific gravity of the solution is about 1.020. Transfer 500 ml. of this solution to a graduated liter flask, dilute to the mark with distilled water, and mix thoroughly. Transfer 25 ml. of this latter solution to a beaker, dilute with distilled water to 300 ml., add a few drops of HNO_3, and then (gradually and with constant stirring to coagulate the precipitate) a 5 per cent solution of $AgNO_3$, until no further precipitation takes place. Heat the liquid to boiling and set the whole aside (in a dark place) to cool. When completely settled, filter off the AgCl on asbestos in a Gooch crucible (previously weighed), wash thoroughly with distilled water containing a few drops of HNO_3 to the liter and then with a little pure water. Dry first at 100° C., and then at 130° to 150° C., cool, and weigh. From the weight of AgCl found calculate the normality of the solution. Instead of the above procedure, the solution may be standardized against the standard NaOH solution (a) that has been standardized against Bureau of Standards standard acid potassium phthalate, using phenolphthalein indicator.

F–3c. *Phenolphthalein indicator.*—Dissolve pure phenolphthalein in 50 per cent ethyl alcohol in the proportion of 5 g. per 1,000 ml.

F–3d. *Standard silver nitrate solution (approximately 0.01 N).*—Dissolve about 1.7 g. of pure silver nitrate in distilled water in a 1-liter volumetric flask, dilute to the mark with distilled water, and mix thoroughly. Standardize volumetrically against pure sodium chloride, using potassium chromate indicator. This solution should be kept in a bottle painted black.

F–3e. *Potassium chromate indicator.*—Dissolve pure, chloride-free potassium chromate (K_2CrO_4) in distilled water, in the proportion of 10 g. per 100 ml.

F–3f. *Silver nitrate solution (10 per cent).*—Dissolve pure silver nitrate in distilled water in the proportion of 10 g. of silver nitrate to 100 ml. of the completed solution.

G. PACKAGING, PACKING, AND MARKING.

G–1. *Packaging.*—Unless otherwise specified, commerical packages are acceptable under this specification.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

G–3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, grade, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

G–4. Any special requirements of the individual departments are noted under section H.

Source: https://assist.dla.mil -- D
Check the source to verify that this

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following departmental specifications of the issue in effect on date of invitation for bids shall, respectively, form a part of this specification in so far as the activities noted are concerned:

H–1a. Army: U. S. Army Specification No. 100–2, Standard Specification for Marking Shipments.

H–1b. Navy: Navy Department General Specifications for Inspection of Material (copies of which may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C.).

H–1c. Marine Corps: Instructions issued by the Quartermaster.

H–2. *Navy Department purchases.*

H–2a. *Packing.*—Unless otherwise specified glycerin shall be delivered in cans containing 5 gallons each, packed 2 cans in a wooden case. Cans and cases shall conform to the issue of Navy Department Specification 65C9 in effect at date of invitation for bids.

H–2b. Navy Department requisitions should state the grade of glycerin required. Grade A is suitable for laboratory work and shall be specified when a lubricant for oxygen compressors or for refrigerating machinery using ethyl chloride is required.

H–2c. Navy Department purchases will be made under the issue in effect on date of invitation for bids, of Navy Department Specification No. 51G1.

I. NOTES.

I–1. Purchasers should specify the grade required. (See pars. E–2c and E–3c.)

I–2. Grade A covers the glycerin known in the trade as "U. S. P." glycerin.

I–3. Grade B is intended for the manufacture of explosives and for other industrial uses where high-gravity glycerin is required.

I–4. Grade C is intended for industrial purposes other than the manufacture of explosives.

I–5. *Basis of purchase.*—Material of each grade shall be purchased by net weight.

I–6. It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–7. This specification governs all United States Government purchases of this commodity. Users are requested to retain this specification in permanent file until a revision is issued.

I–8. Interested parties are advised that an Alphabetical Index of Federal Specifications may be obtained upon application as noted in paragraph next below, at a cost of 5 cents.

Source: https://assistdla.mil - Check the source to verify that

I–9. Copies of this specification may be obtained upon application, accompanied by money order or coupon, or cash, to Superintendent of Documents, Government Printing Office, Washington, D. C. Price, 5 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded, by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.m
Check the source to verify th